Physics-Informed Machine Learning for Multi-Second Aircraft Stall Early Warning and Cross-Mechanism Generalization

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Abstract — Aerodynamic stall is preceded in principle by a measurable approach to the critical angle of attack, but whether that approach leaves a *learnable, multi-second* signature depends on how the aircraft is flown into it. We investigate this inside a self-contained simulation study: a 2-D longitudinal point-mass flight-dynamics model with an emergent, smooth $C_L(\alpha)$ stall, three generations of synthetic trajectory datasets, and a temporal machine-learning early-warning model. The first evaluated dataset generation (v0.2; 1,000 trajectories, 1,753,615 rows) gave a median *physical* precursor — the direction-aligned transit from α = 8° to the 16.07° stall boundary — of only 0.54 s, and a model trained on it behaved accordingly: 100% event recall but a 0.53 s median credited lead time, with no event warned ≥4 s ahead. A physics diagnosis attributed the short precursor to control-input timing rather than to the dynamics. Leaving the physics engine untouched and re-timing only the elevator profile produced v0.3 (3,150 trajectories, 5,340,865 rows), median physical precursor 4.38 s; the same frozen model family then reached PR-AUC 0.890, 96.1% event recall (evaluated events) and a 4.72 s median lead among warned events. A zero-exposure regime-exclusion control collapsed to PR-AUC 0.552 and 0.73 s median lead, so we tested transfer to a structurally distinct single-pulse mechanism: the frozen model retained PR-AUC 0.835 with 100% event recall (46/46) and a 2.96 s median lead. The evidence supports family-level generalization across control-input mechanisms producing the same slow approach to the boundary — not universal zero-shot stall prediction, and not a claim about any real aircraft.

Index terms — flight dynamics, aerodynamic stall, early warning, time-series classification, generalization, simulation.

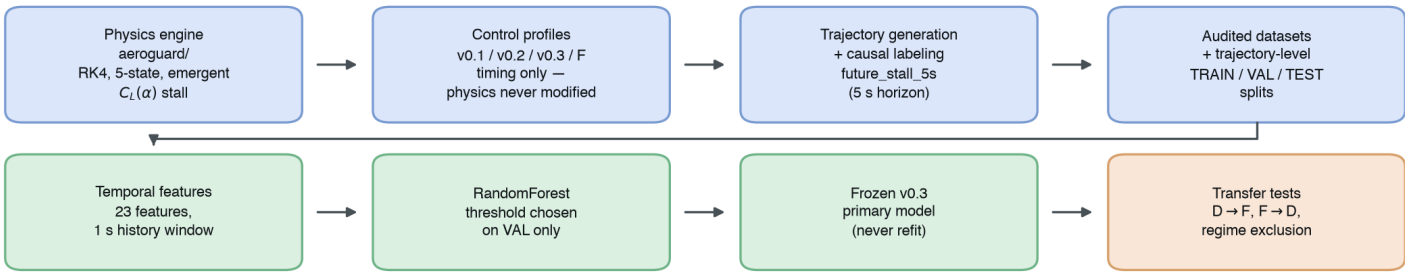


Figure 1. AeroGuard end-to-end architecture. The physics engine (top left) is fixed after Stage 1; every subsequent dataset generation changes only the control-input profile. The v0.3 primary model is frozen before any transfer test. Compact redraw of the schematic at `outputs/final/figures/01_aeroguard_pipeline.png` — same stages, same flow [A1].

1. Introduction

Aerodynamic stall occurs when the angle of attack α exceeds the value at which the lift coefficient peaks, after which lift falls with further increases in α [1]. Operationally the hazard is not the stall itself but how little time separates a recoverable state from an unrecoverable one. Conventional stall-warning devices are by construction *imminent-event detectors*: they fire when a measured quantity crosses a fixed margin ahead of the critical angle [10]. A system that instead recognised the multi-second *approach* would sit closer to the early-warning-signal literature on slow approaches to critical transitions [2] than to threshold alarms.

Whether such a system is learnable is contingent on a fact that is easy to overlook: **the precursor must exist in the data before a model can learn it.** A trajectory commanded abruptly through the boundary carries no multi-second signature, whatever model is applied — and that contingency became this investigation's central finding rather than a preliminary detail.

We ask two questions. First: *can a physics-based flight simulator produce trajectories with a genuine, multi-second precursor to aerodynamic stall, and — if so — can a machine-learning model learn to detect that precursor early enough to constitute a useful early-warning system, rather than an imminent-event detector reacting inside the final half-second before the event?* Second, reachable only once the first is answered affirmatively: *is the learned skill a transferable understanding of rising α with shrinking stall margin over several seconds, or a memorized fingerprint of one control-input shape?*

The contributions are a pipeline that measures the physical precursor duration directly rather than assuming it; a diagnosis attributing an early dataset's sub-second precursor to control-profile timing rather than to the dynamics, with an intervention that changed only the control input; and a two-direction transfer test that separates *family-level* generalization from the zero-shot claim the headline numbers might otherwise invite. All of it is simulation-internal: no real flight, wind-tunnel or validated-airframe data was used, and no claim about real aircraft is made or implied.

2. Methods

2.1 Flight-Dynamics Model

The simulator (`aeroguard/`) integrates a 2-D longitudinal point-mass model with the five-element state $\mathbf{x} = [V, \gamma, \theta, h, q]$ — airspeed, flight-path angle, pitch angle, altitude, pitch rate — by fixed-step fourth-order Runge–Kutta at $\Delta t = 0.01$ s over a 20 s horizon. Angle of attack is derived, not stored: $\alpha = \theta - \gamma$. The equations of motion take the standard longitudinal form [3], [4]:

$$m\,\mathrm{d}V/\mathrm{d}t = T\cos\alpha - D - m\,g\sin\gamma \quad (1)$$

$$m\,V\,\mathrm{d}\gamma/\mathrm{d}t = L + T\sin\alpha - m\,g\cos\gamma \quad (2)$$

$$\mathrm{d}h/\mathrm{d}t = V\sin\gamma,\ \mathrm{d}\theta/\mathrm{d}t = q \quad (3)$$

$$I_{yy}\,\mathrm{d}q/\mathrm{d}t = M_\delta\,\delta_e - M_q\,q - M_\alpha\,\alpha \quad (4)$$

Equation (4) is a **linear short-period surrogate** for the pitching-moment equation, not a full aerodynamic moment model: the elevator has authority over pitch rate, damped by q and weakly restored by α. Drag follows the polar $C_D = C_{D0} + k\,C_L^2$; thrust is linear in throttle.

Stall is emergent, not rule-based. The lift coefficient is a sigmoid blend — the technique standard for small fixed-wing UAV models [5] — between a linear pre-stall curve $C_L = C_{L0} + C_{L\alpha}\,\alpha$ and a post-stall branch decaying exponentially from that curve's value at α_{stall}; no `if alpha > threshold`

branch exists anywhere in the aerodynamics module. The **stall boundary** used for labeling is therefore not a hand-set constant but the numerically located argmax of the live $C_L(\alpha)$ function, $\alpha^* \approx 16.07°$. Air density is constant at sea level ($\rho = 1.225$ kg m^{-3}), as is mass; the coefficients ($m = 1200$ kg, $S = 16.2$ m², $I_{yy} = 1285$ kg m²) are plausible defaults for a generic small fixed-wing aircraft, **not** values measured or fitted for any real airframe. Critically for what follows, the physics engine was frozen after Stage 1 and never modified again.

2.2 Dataset Generation and Labeling

Each dataset version simulates trajectories under randomized initial conditions and regime-specific elevator/throttle profiles. v0.2 comprises 1,000 trajectories (500 `normal`, 250 `stall`, 250 `near_boundary`; 1,753,615 rows; 19.2% crossing the boundary); v0.3 comprises 3,150 (500 `normal`, 250 `stall`, 2,400 `gradual_approach_v3`; 5,340,865 rows; 470 crossings), the last count sized from the Wilson 95% lower bound on the calibrated crossing rate so as to guarantee at least v0.2's crossing count.

Per-timestep features are the state and controls, causal one-step derivatives, and `stall_margin` = $\alpha^* - \alpha$. The target is `future_stall_5s`: 1 if $|\alpha|$ exceeds α^* anywhere in the half-open window $(t, t + 5$ s], 0 if not, undefined where that window is incomplete — strictly causal, never inspecting the current row. Splitting is at the **trajectory** level with verified zero trajectory-ID overlap across train, validation and test, and integrity audits (no NaN/Inf, no duplicate IDs, monotonic timestamps, causal-derivative re-derivation, no sub-zero altitudes) passed at every version.

2.3 Machine-Learning Method

The primary model is a random forest [6] built with scikit-learn [7] over 23 features at a 1 s history window: 8 instantaneous state/control channels (V, α, γ, q, h, δ_e, throttle, `stall_margin`), 5 causal derivatives, and 10 windowed summaries of the preceding second (α mean/min/max/range/slope/trend, and V, γ, q slopes plus elevator change). Hyperparameters, feature definitions, threshold selection and the event definition were fixed on v0.2 and reused **unchanged** for v0.3, the regime-exclusion control and the cross-mechanism experiment. The threshold is chosen on validation only; the test split is scored once.

2.4 Evaluation Protocol

Because positives are rare and the negative class dominates, we report **PR-AUC** as the primary row-level metric, precision–recall curves being more informative than ROC curves under class imbalance [8], [9]. Row-level metrics alone cannot support an early-warning claim, so we add two event-level measures: a discrete stall **event** counts as warned if an alarm is raised anywhere in the 5 s label horizon preceding the crossing, the **credited lead time** is the earliest such alarm's distance to the crossing (capped at that horizon), and **warning coverage at τ** is the fraction of events warned at least τ seconds ahead. Events are counted only within the evaluated rows, which require an available label (the last 5 s of each trajectory has no complete future window) and a full feature window. A crossing in the last 5 s of a trajectory is therefore outside the evaluation; it is neither warned nor missed. The median credited lead time is taken over warned events only, and missed events carry no lead time. An event is a maximal run of unsafe samples, so a trajectory that re-enters the unsafe region contributes more than one event.

Separately from the model, we measure each crossing's **physical precursor duration** from the trajectory itself: the direction-aligned, dip-aware time from an $\alpha = 8°$ onset to the crossing. This model-independent metric makes the diagnosis of §3.2 possible, since it bounds what any model could achieve on a given dataset.

3. Experimental Design and Results

3.1 v0.2: The Short-Precursor Problem

v0.1 (1,000 trajectories, 1,565,280 rows) established the generation machinery; v0.2 regenerated it with a recalibrated near-boundary regime and explicit ground-contact termination, and is the first version evaluated end to end. An instantaneous-state baseline on v0.2 set the reference point: a random forest on point-in-time features reached test PR-AUC 0.742 (precision 0.814, recall 0.604), against 0.599 for a tuned α-threshold rule. Adding temporal context improved row-level performance to PR-AUC 0.813 — but the early-warning picture told a different story. Row-level recall was 98.8% within 0.5 s of the crossing and then fell off a cliff, never exceeding 26% in any later bucket and reaching 0.0% at 4–5 s (Fig. 3a). All 14 discrete crossings in the usable test population were warned, but the **median credited lead time was 0.53 s** (Table 1). By the definition adopted in §1, this is an imminent-event detector, not an early-warning system.

3.2 Physics-Based Precursor Diagnosis

We next asked whether this ceiling was a property of the model or of the data. A direction-aligned analysis of the 5 s preceding every v0.2 crossing (Fig. 2a) showed median α essentially flat from $t - 5$ s to about $t - 1$ s in both regimes that ever cross, rising sharply only inside the final second: the median $8° \rightarrow 16°$ transit was 0.54 s in `near_boundary` and 0.33 s in `stall`. Single-variable separability between soon-to-cross and safe rows sat near chance (AUC 0.51–0.65) at every lead time beyond ~1 s in `stall`, and `near_boundary`'s higher separability traced to a *level* effect — crossing trajectories start at higher α — not a *trend* effect. Physical precursor coverage was ~4% at ≥2 s and 0% at ≥3 s (Fig. 2b).

The conclusion was therefore not "the model is too weak" but "**the precursor is not in the data**." The proximate cause was v0.2's large-magnitude, long-hold elevator pulses, aimed at an equilibrium α far past the boundary, which drove a fast transit through the precursor region — a control-profile timing artifact, not evidence that the dynamics cannot support a slower approach.

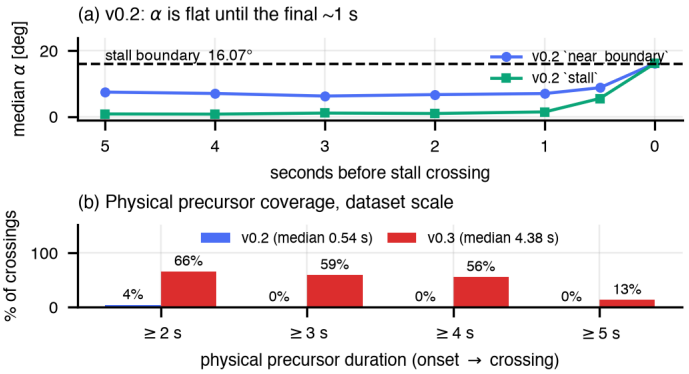


Figure 2. The precursor is a property of the control profile, not of the physics. **(a)** Direction-aligned median α in the 5 s before a v0.2 stall crossing, by regime: α is flat until roughly the final second in both. **(b)** Physical precursor coverage at dataset scale for v0.2 (median 0.54 s) versus v0.3 (median 4.38 s, n = 194 usable crossings). Composed from recorded outputs only [A2], [A3]; no experiment was re-run.

3.3 v0.3 Control-Profile Intervention

The intervention is deliberately narrow: **the physics engine, the aircraft parameters, the $C_L(\alpha)$ curve and the stall boundary were not modified**, nor were the features, the model family, the label definition or the evaluation procedure. Only the elevator input's timing and shape changed, introducing a new `gradual_approach_v3` regime.

That asymmetry is what makes the change a test of the hypothesis rather than a way of making the task easier. The classification problem is not relaxed — boundary, label horizon and metrics are identical, and a slow approach spends *longer* in the ambiguous region near the boundary — so what changes is only whether a multi-second precursor is physically present to detect. If §3.2's diagnosis holds, re-timing should lengthen the measured physical precursor and the model's lead time should follow; if the sub-second ceiling were intrinsic to the dynamics, re-timing would move neither.

The first attempt failed informatively: five single-pulse candidates with lengthened rise times only tripled the median transit to 0.90–1.76 s while flight-path-angle envelope terminations rose from 22.7% to 57–80%, and **none produced a single ≥3 s precursor event**. A two-stage, two-pulse design ("Candidate D") did far better, though its first-reported ≥3 s rate proved inflated ≈2.5× by a metric crediting dive-then-zoom recovery as precursor time (corrected: 33.3%). Two further rounds fixed the remainder: same-sign, zero-gap pulse sequencing eliminated dive/zoom-climb crossings (0/40), and a 7.0 s pulse-duration cap trimming hold time only cut non-crossing γ-termination to 28.6%, below the original 31.4% baseline.

At full scale, v0.3 delivered 194 usable crossings with a **median physical precursor of 4.38 s**: 66.0% ≥2 s, 59.3% ≥3 s, 55.7% ≥4 s. Physical classification of the 317 `gradual_approach_v3` crossings was 99.1% clean gradual/monotonic low-γ approaches, 0.9% runaway, and 0% dive-then-zoom.

3.4 Temporal ML Results

Re-running the *unchanged* v0.2 temporal procedure on v0.3 (538,034 evaluated test rows from 456 of the 473 test trajectories) produced PR-AUC **0.890** (precision 0.935, recall 0.736), event recall **96.1% (73/76 evaluated events)** and a **median credited lead time of 4.72 s among the 73 warned events** — roughly nine times v0.2's. Warning coverage at ≥4 s rose from 0.0% to 55.3%, and row-level recall in the 4–5 s bucket from 0.0% to 48.4% (Fig. 3, Table 1). The 76 evaluated events are 57 first crossings and 19 second episodes in trajectories that re-enter the unsafe region. Ten further test crossings (8 `stall`, 2 `gradual_approach_v3`), each the first crossing of a different trajectory, fall in the last 5 s of their trajectories and are outside the evaluation. Among the 57 first crossings, 54 were warned.

Two observations temper this. First, the false-positive *character* changed: in v0.2 every false positive lay on a trajectory that never stalls, whereas in v0.3 70% of unique false-positive trajectories do cross the boundary elsewhere in their own telemetry — ambiguous early rows on genuine approach shapes. Second, the population-median ML lead time (4.90 s) tracks the median physical precursor (4.39 s) closely, but the *per-event* correlation is only moderate (Pearson r = 0.337, n = 22): the model behaves more like "this resembles a gradual approach, warn now" than a per-trajectory timer.

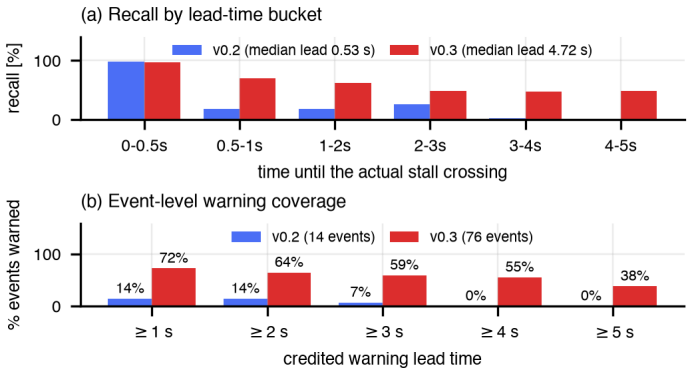


Figure 3. Temporal early-warning performance, v0.2 versus v0.3, same model family and evaluation procedure throughout. **(a)** Row-level recall by time-to-crossing. **(b)** Event-level warning coverage at increasing lead-time thresholds. Composed from the two experiments' recorded metrics [A2], [A4].

4. Cross-Mechanism Generalization

4.1 Regime-Exclusion Experiment

A model that has learned only "the `gradual_approach_v3` shape" and one that has learned "α rising toward a shrinking margin" are indistinguishable in distribution. Our first probe was a regime-exclusion control: retrain the identical model with **all 1,691** `gradual_approach_v3` training trajectories removed, then evaluate unchanged on the full test split.

The multi-second capability collapsed. PR-AUC fell 0.890 → **0.552**, event recall 96.1% → **64.5%**, median credited lead time 4.72 s → **0.73 s**, coverage at ≥4 s 55.3% → **11.8%**. Near-immediate (0–0.5 s) detection degraded only modestly, 97.2% → 81.2%, but every bucket beyond ~0.5 s fell by a factor of 20–40 (Fig. 4a) — a dataset-shift failure [11] consistent with a meaningful share of the headline capability being regime-shape memorization.

4.2 Alternative Mechanism (Candidate F)

The exclusion control answers a blunt question — "can the model learn a multi-second precursor from *zero* exposure to any slow approach?" (no) — but cannot separate memorizing Candidate D's two-pulse *staircase* from learning something transferable. We therefore built a structurally distinct mechanism, "**Candidate F**": a **single**, duration-capped elevator pulse (cap 6.0 s) giving one smooth monotonic α rise to a single plateau, against Candidate D's staircase of rise → partial fall → rise → fall. Both reach the same physics-defined boundary through the same aircraft and pitch model by a different temporal path, and the new module touches neither the physics engine, the shared configuration, nor either Candidate D module. Its 150-trajectory calibration passed cleanly: 0% dive-then-zoom, 0% runaway, 100% clean crossings, 100% ≥2 s and 62.5% ≥3 s precursor coverage.

4.3 Forward and Reverse Transfer

Forward. The frozen v0.3 model — never refit — was evaluated on 259,025 usable rows from 293 held-out Candidate F trajectories (46 events), none seen during training in any form. It retained PR-AUC **0.835**, 94% of its in-distribution value, with **100% event recall (46/46)** and a **2.96 s** median credited lead time; ≥2 s coverage, at 89.1%, exceeds the in-distribution figure itself.

Reverse. One fresh forest, trained only on Candidate F trajectories and never shown a single `gradual_approach_v3` example, was evaluated on the frozen v0.3 test split's held-out `gradual_approach_v3` rows: PR-AUC **0.708**, event recall **87.0% (47/54)**, median credited lead time **5.00 s** (at the horizon cap).

Table 1 places all five evaluation conditions side by side.

Table 1. The five evaluation conditions. "Cov. ≥τ" is the fraction of stall events warned at least τ s before the crossing; lead times are capped at the 5 s label horizon. All values reproduced from the frozen experiment reports [A2], [A4], [A5]. Event recall and median lead time are over the evaluated events (see §2.4); the median lead time is over warned events only.

Condition	PR-AUC	Event recall	Lead (s)	≥2 s	≥4 s
v0.2 temporal	0.813	100.0% (14/14)	0.53	14.3%	0.0%
v0.3 in-distribution	**0.890**	96.1% (73/76)	**4.72**	64.5%	**55.3%**
Regime exclusion	0.552	64.5% (49/76)	0.73	22.4%	11.8%
Forward D → F	0.835	**100%** (46/46)	2.96	**89.1%**	37.0%
Reverse F → D	0.708	87.0% (47/54)	5.00	72.2%	46.3%

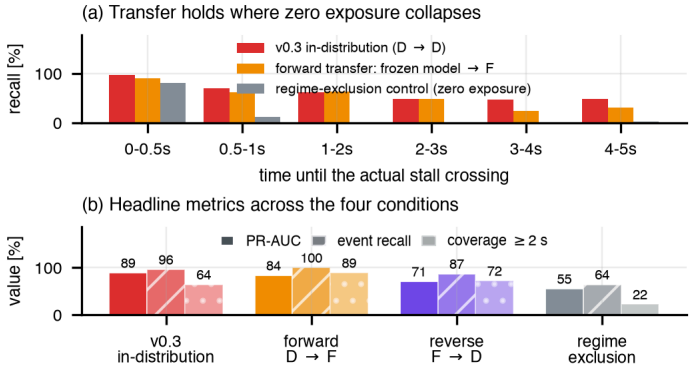


Figure 4. Cross-mechanism generalization. **(a)** Row-level recall by time-to-crossing for the in-distribution model, the same frozen model on the novel Candidate F mechanism, and the zero-exposure regime-exclusion control. **(b)** Headline metrics across all four learned conditions. Transfer to a structurally distinct mechanism preserves multi-second performance; removing the phenomenon class from training does not. Composed from recorded outputs [A2], [A5].

5. Discussion

Two capabilities must be kept apart. The first is **multi-second early warning**: on v0.3 the model warns 96.1% of events at a 4.72 s median lead against 0.53 s on v0.2, under an identical procedure. Because only the control-input timing differed, the gain is attributable to a physical precursor now being present in the data, not to a stronger model or a relaxed task (§3.3).

The second is **generalization**, where the two probes answer different questions and must not be conflated. The zero-exposure check (§4.1) shows

the capability is not inferred from the dynamics: a model that has never seen a slow approach does not derive one from "α rising, margin shrinking," and its lead time falls back to the sub-second regime (0.73 s). Removing the phenomenon class removes the skill. Against that, the frozen model transfers to Candidate F — whose α trajectory is a single smooth hump, not a staircase — retaining 94% of in-distribution PR-AUC and warning 46 of 46 events, with the reverse direction also holding (PR-AUC 0.708, 87.0% recall).

These bound the claim on both sides. What the model learned is not a memorized fingerprint of one control-input shape, since a structurally distinct shape transfers; but neither is it a shape-independent grasp of approach-to-stall physics, since zero exposure collapses it. The defensible description is **family-level generalization** across realizations of a slow α-rise toward the boundary — **not zero-shot generalization** to arbitrary unseen regimes, which this work neither tests nor claims.

6. Limitations and Threats to Validity

Simulation only. The model is a simplified 2-D longitudinal point-mass simulator: no roll, yaw or sideslip, constant sea-level air density, a linear pitch-response surrogate (Eq. 4), a linear throttle-to-thrust map. It is **not a validated model of any real aircraft** — every coefficient is a plausible default, not measured or fitted data — and nothing here supports a claim about real-aircraft behaviour or about deployment in any aviation system.

Family-level, not zero-shot, generalization. §4.3's transfer required prior exposure to some slow-approach mechanism; §4.1 shows the capability does not emerge without it. Candidate F, though structurally distinct from Candidate D, reuses an already-calibrated elevator specification and the same duration-cap concept, so it tests transfer *within* the broad slow, elevator-driven approach family — not to a qualitatively different stall-inducing mechanism (turbulence-induced, asymmetric or non-elevator-driven), none of which this simulator models.

The `stall` regime remains an imminent-only detector. Its physical precursor is ≈0.3 s in both dataset versions and recall beyond ~1 s is 0% in both; nothing here extended its warning time.

Weak per-event timing calibration. Population-median lead time tracks the population-median physical precursor closely, but the per-event correlation is only $r = 0.337$ ($n = 22$); warning times are not per-trajectory time-to-stall estimates.

Small absolute event counts. 76 events in the v0.3 primary test population, 46 and 54 in the two transfer directions. The percentages support the qualitative conclusions but carry real sampling uncertainty; small differences between conditions should not be over-read. The evaluated events exclude ten test crossings that occur in the last 5 s of their trajectories, mostly in the `stall` regime (8 of 25 `stall` crossings versus 2 of 42 `gradual_approach_v3`); the effect of these events on recall was not measured.

Limited regime coverage. Only three broad regimes (`normal`, `stall`, slow approach) and one aircraft configuration were simulated; behaviour outside them is untested.

Process threats. Conflicting v0.3 calibration results arose partway through the project and were reconciled with a corrected, direction-aligned precursor metric; this paper cites the reconciled figures throughout [A1].

7. Conclusion

Answering §1 directly: within this simulator a multi-second stall precursor is achievable through control-input timing alone (median 0.54 s → 4.38 s), a standard temporal model learns to exploit it (96.1% event recall over evaluated events, 4.72 s median lead among warned events), and that skill transfers in both directions to a structurally distinct control mechanism — but does not survive removal of the phenomenon class from training. Stated precisely: **the evidence supports multi-second stall early warning and transfer across structurally distinct control-input mechanisms producing the same underlying physical phenomenon, but does not establish universal zero-shot stall prediction across arbitrary unseen flight regimes.**

The methodological takeaway cost the most to learn: the dominant constraint on lead time was a property of the *data-generating process*, measurable independently of any model, and measuring it first turned an apparent modelling ceiling into a dataset-design problem. Natural extensions, none pursued here, are transfer to a non-elevator-driven stall mechanism, better per-event lead-time calibration, and dynamics beyond 2-D longitudinal motion — a prerequisite before any real-aircraft-relevant claim.

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Primary sources (this repository). Every AeroGuard-specific number above is reproduced from these frozen artifacts; none was recomputed for this paper. **[A1]** *AeroGuard — Final Research Report* (frozen), `outputs/final/`, with `PROVENANCE.md`, `REPRODUCIBILITY.md`. **[A2]** v0.2 temporal experiment, `outputs/ml_temporal/`. **[A3]** Precursor diagnosis, v0.3 generation, audit and calibration: `outputs/precursor_diagnosis/`, `outputs/dataset_audit_v3/`, `outputs/v03_calibration/`. **[A4]** v0.3 temporal ML and regime exclusion, `outputs/ml_v03/`. **[A5]** Cross-mechanism transfer, `outputs/ml_v03_generalization/`.

Published on the AeroGuard project website. **Not peer-reviewed, not flight-tested, and not a certified aviation safety tool.** All results are internal to the simulator described in §2.1.